

Exercise 1: System Hardware Detective

- Processor (CPU): 2.3 GHz 8-Core Intel Core i9
- Total RAM: 32GB (2667 MHz DDR4)
- Storage Device: SSD (PCI Express Drive)
 - Capacity: 1TB total (Macintosh HD: 855 GB, BOOTCAMP: 135.24 GB)
- OS Version: macOS Tahoe 26.6.2 (MacBook Pro 16inch, 2019)

Analysis Questions

Q: Based on your CPU cores, how many tasks can your processor theoretically handle simultaneously?

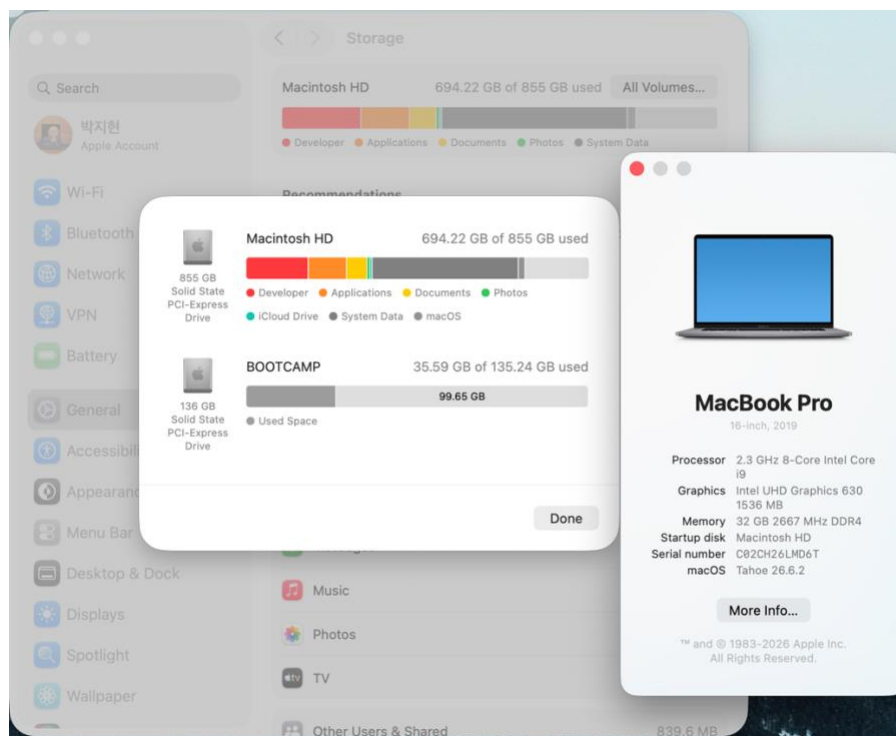
A: My CPU has 8 physical cores, so it can theoretically handle 8 tasks simultaneously, with each core handling one task at a time.

Q: Is your storage primarily HDD or SSD? What are the performance implications?

A: My laptop use SSD. SSD is much faster than HDD because no moving parts inside. So boot up is fast and open app fast.

Q: How does your RAM amount compare to typical requirements for modern applications?

A: I have 32GB RAM, so this is more than normal apps need. I can run many apps at the same time without slowing down.



Exercise 2: Operating System Feature Hunt

1. Process Management

- 5 Running Processes: WindowServer, Activity Monitor, ContinuityCaptureAgent, kernel_task, screencapture
- Process using the most CPU: WindowServer (44.3%)

2. Memory Management

- Total RAM Usage: Around 22 GB used from 32 GB
- Application using the most memory: WindowServer (about 1 GB)
- Available Free Memory: Around 10 GB left

3. File System Management

- Navigated Folder: Documents
- Created Folder: OS_Lab_Test
- Creation Date: Sep 9, 2026, 10:02 PM

4. Device Management

- 3 Hardware Devices: Screen, Camera, Wi-Fi
- Device Driver: Intel Graphics Driver

5. Security Features

- Antivirus Status: Mac basic security (XProtect) is running in background.
- Automatic Updates: It is turn on in settings.
- Recent Security Scan: Mac check bad files automatically.

Challenge Questions

Q: Which process is using the most resources and why might that be?

A: WindowServer. It draws things on the screen, like windows and icons, so it works all the time.

Q: What would happen if your OS didn't manage memory automatically?

A: Apps could use too much memory and the computer could become very slow or freeze.

Q: How does your OS protect you from security threats?

A: Mac check downloaded file before open, block virus, and keep update system.

Activity Monitor

All Processes

ⓧ ⓘ ⋮

CPU

Memory

Energy

Disk

Network

🔍

Process Name	% CPU	CPU Time	Threads	Idle Wake Ups	% GPU	GPU Time	PID	User
WindowServer	44.3	21:09.22	16	98	5.7	5:01.11	217	_windowserver
Activity Monitor	12.2	4:51.37	18	4	0.0	0.00	2654	jihyun
ContinuityCaptureAgent	3.9	1:01.25	8	3	0.0	0.00	481	jihyun
kernel_task	3.8	7:23.31	365	747	0.0	0.00	0	root
screencapture	3.2	0.48	3		0.0	0.00	8753	jihyun

Activity Monitor

All Processes

ⓧ ⓘ ⋮

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🔍

Process Name	Mem...	Threads	Ports	PID	User
WindowServer	1.06 GB	15	4,430	217	_windowserver
Code Helper (Plugin)	382.4 MB	39	151	3680	jihyun
Notion Helper (Renderer)	380.7 MB	21	318	1358	jihyun
https://learn.senecapolytechnic.ca	325.1 MB	8	73	3469	jihyun
Microsoft Word	260.4 MB	16	822	7799	jihyun

MEMORY PRESSURE

Physical Memory:	32.00 GB	App Memory:	14.31 GB
Memory Used:	21.85 GB	Wired Memory:	4.72 GB
Cached Files:	9.77 GB	Compressed:	1.58 GB
Swap Used:	0 bytes		

OS_Lab_Test Info

OS_Lab_Test

Zero KB

Modified: Today, 10:02 PM

Add Tags...

General:

Kind: Folder
Size: Zero bytes (Zero bytes on disk) for 1 item
Where: iCloud Drive > Documents
Created: September 9, 2026 at 10:02 PM
Modified: September 9, 2026 at 10:02 PM

Shared folder

Locked

More Info:

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Name & Extension:

OS_Lab_Test

Hide extension

Comments:

Preview:

Sharing & Permissions:

You can read and write

Name	Privilege
jihyun (Me)	Read & Write
staff	Read only
everyone	Read only